

Tony Kabilan Okeke

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Education

Drexel University, Philadelphia, PA Sep 2019 – Jun 2024
M.S., Biomedical Engineering, 2024. Concentrations: Bioinformatics and Bioimaging. GPA: 3.99/4.00.
B.S., Biomedical Engineering, 2024. Concentration: Neuroengineering. GPA: 3.99/4.00.

Research and engineering experience

Machine Learning Engineer Aug 2024 – Present
Moberg Analytics, Philadelphia, PA

- Led model development, experiments, cross-validation, testing, manuscript drafting, and peer-review revisions for a first-author PRECICECAP study on arterial blood pressure artifact detection, published in *Biomedical Signal Processing and Control*. Released ABP and related ECG artifact-detection models and inference code through OkekeClean.
- Implemented density spectral array (DSA) processing for ELECTRO-BOOST, an ancillary study to BOOST-3. Processed over 1,200 EEG recordings from over 200 patients; 10 clinicians use the DSA alongside raw EEG in the study's change-point review workflow.
- Built offline and on-demand DSA pipelines that preserve causal filter behavior when processing arbitrary recording segments. The feature is available at at least three hospitals in the US and Europe, in addition to the study environment.
- Developed a custom EDF parser after benchmarking existing libraries against memory and loading-time requirements. It supports access to over 1,100 EEG files in ELECTRO-BOOST's shared review platform.
- Developed Moberg's ACME mobile application and sensor integrations for the US Army TATRC AutoDoc program, funded through MTEC. Rebuilt a React Native prototype in native Kotlin to support concurrent Bluetooth sensor streams and device SDKs. Assembled five kits; delivered four for TATRC evaluation and retained one internally.
- Own EEG paging and data-loading improvements for continuous playback in a cloud environment. Support customer deployments, diagnose failures, and deliver hotfixes; contribute requirements and QMS documentation toward a planned 510(k) submission.
- Built an early CLARA AI prototype for Moberg's successful Ben Franklin Technology Partners funding pitch and participated in technical Q&A. Currently developing a research-agent demonstration for an upcoming NCS informatics workshop.

Data Scientist Apr – Sep 2023
IVIVT Non-Clinical Safety, GSK, Collegeville, PA

- Developed quality-control, visualization, and cell-type annotation workflows for 10X Visium spatial transcriptomics; built an interactive GeoMx analysis platform.
- Worked with the HESI eSTAR carcinogenicity group on neural-network models of molecular initiating events using rat transcriptomics data, and explored compound embeddings and biomarker models using ChEMBL data.

Research and engineering experience, continued

Data Scientist

Apr – Sep 2022

IVIVT Non-Clinical Safety, GSK, Collegeville, PA

- Built a Dash/Flask application for statistical analysis and biomarker discovery in high-content imaging data; used Dask for computation and deployed through RStudio Connect.
- Implemented decision-tree, random-forest, and SVM models for biomarker discovery, and integrated Noise2Void and Cellpose into image-processing workflows.

Computational Research Assistant

Mar 2021 – Aug 2022

Invenio Lab, Penn Medicine, Philadelphia, PA

- Analyzed longitudinal serum and urinary biomarkers in COVID-19 and cardiac-surgery cohorts, contributing to publications on immune activation, neuroinjury, and complement pathways.
- Developed Python and R workflows for ELISA and sequencing data ingestion, quality control, and analysis, including differential methylation and pathway enrichment.

Research Assistant

May 2020 – Jun 2021

Zhou Lab, Children's Hospital of Philadelphia, Philadelphia, PA

- Contributed to SeSAmE, an R package for Infinium DNA methylation analysis, and developed tests against public Gene Expression Omnibus datasets.

Selected projects

OkekeClean

2026

Python; physiological waveform artifact detection

- Released ABP and ECG model weights with inference code and documentation. Source code; ABP model; ECG model.

Meddibia

2023 – 2024

Team hackathon and prototype project

- Built an AI health-assistant prototype for Philly Codefest 2023 and later explored medical-record digitization through customer discovery and an EMR prototype. The project did not reach customer deployment; prototype work ended in December 2024.

WazobiaCode

May – Sep 2024

FastAPI, Next.js; education project

- Organized a web-development bootcamp for over 100 Nigerian high-school students and built a learning platform with progress tracking and Telegram integration.

MLGO: Gene Ontology enrichment prediction

2022 – 2024

Python, R, PyTorch, limma

- Built a preprocessing and differential-expression pipeline for over 11,000 RNA-seq datasets from DEE2. Trained autoencoders and explored learned representations for Gene Ontology term prediction; presented at the 2024 Society of Toxicology meeting.

Peer-reviewed journal articles

1. **Okeke, T. K.**, Shrestha, M., Moyer, E., Hirsch, K. G., May, T. L., He, Z., Tam, J., Faiver, L., Moberg, R., and Elmer, J. (2026). Evaluating artifact detection algorithms for the arterial blood pressure waveform acquired from the intensive care unit: A PRECICECAP informatics approach. *Biomedical Signal Processing and Control*, 124, 110434. doi:10.1016/j.bspc.2026.110434.
2. Laudanski, K., Liu, D., **Okeke, T.**, Restrepo, M., and Szeto, W. Y. (2022). Persistent Depletion of Neuroprotective Factors Accompanies Neuroinflammatory, Neurodegenerative, and Vascular Remodeling Spectra in Serum Three Months after Non-Emergent Cardiac Surgery. *Biomedicines*, 10(10), 2364. doi:10.3390/biomedicines10102364.
3. Laudanski, K., Liu, D., Gullipalli, D., Song, W. C., **Okeke, T.**, and Szeto, W. Y. (2022). A decline of protective apolipoprotein J and complement factor H concomitant with increase in C5a 3 months after cardiac surgery—Evidence of long-term complement perturbations. *Frontiers in Cardiovascular Medicine*, 9, 983617. doi:10.3389/fcvm.2022.983617.
4. Laudanski, K., **Okeke, T.**, Siddiq, K., Hajj, J., Restrepo, M., Gullipalli, D., and Song, W. C. (2022). A disturbed balance between blood complement protective factors (FH, ApoE) and common pathway effectors (C5a, TCC) in acute COVID-19 and during convalesce. *Scientific Reports*, 12, 13658. doi:10.1038/s41598-022-17011-7.
5. Laudanski, K., Hajj, J., Restrepo, M., Siddiq, K., **Okeke, T.**, and Rader, D. J. (2021). Dynamic Changes in Central and Peripheral Neuro-Injury vs. Neuroprotective Serum Markers in COVID-19 Are Modulated by Different Types of Anti-Viral Treatments but Do Not Affect the Incidence of Late and Early Strokes. *Biomedicines*, 9(12), 1791. doi:10.3390/biomedicines9121791.
6. Laudanski, K., **Okeke, T.**, Hajj, J., Siddiq, K., Rader, D. J., Wu, J., and Susztak, K. (2021). Longitudinal urinary biomarkers of immunological activation in covid-19 patients without clinically apparent kidney disease versus acute and chronic failure. *Scientific Reports*, 11, 19675. doi:10.1038/s41598-021-99102-5.
7. Laudanski, K., Hajj, J., Antallosky, B., Ghani, D., Phan, U., Hernandez, R., **Okeke, T.**, Wu, J., Rader, D., and Susztak, K. (2021). Unbiased Analysis of Temporal Changes in Immune Serum Markers in Acute COVID-19 Infection With Emphasis on Organ Failure, Anti-Viral Treatment, and Demographic Characteristics. *Frontiers in Immunology*, 12, 650465. doi:10.3389/fimmu.2021.650465.

Journal peer review

Biomedical Signal Processing and Control, Elsevier, 2025 – present.

Four completed review assignments across three manuscripts, including one re-review; one additional manuscript review in progress. Completed reviews recorded on ORCID.

Technical skills

Languages: Python, R, Bash, C++, SQL, JavaScript/TypeScript, Kotlin.

Machine learning: PyTorch, Transformers, scikit-learn, Weights & Biases.

Software and analysis: FastAPI, Flask, React, Android, Bluetooth sensor integration, Docker, physiological signal processing, Seurat, SquidPy, CellProfiler.

Conference abstracts and presentations

- **Okeke, T. K.**, et al. (2026). Spectrogram-Based Transfer Learning for Electrocardiogram (ECG) Artifact Detection in the ICU. *Critical Care Medicine*, 54(3S), abstract 1599. doi:10.1097/01.ccm.0001188392.79833.43. Presented at SCCM Research Snapshot Theater, Chicago, March 24.
- **Okeke, T. K.**, et al. (2025). Transfer Learning for Artifact Detection in ICU-Collected Arterial Blood Pressure Waveforms. Neurocritical Care Society Annual Meeting, Montreal, September 19. Selected for Distinguished Professor Rounds; presented by coauthor Ethan Moyer.
- **Okeke, T. K.**, et al. (2025). Development of the Autonomous Communications Medical Ecosystem (ACME): A Multimodal Sensor Suite for Passive Data Collection During Combat Casualty Care. Poster MHSRS-25-15162, Military Health System Research Symposium, Kissimmee, FL, August 5. Presenting author.
- Moyer, E., Grym, G., **Okeke, T. K.**, Vitkovitsky, C., Youssef, A., Kim, E., Petrov, D., and Moberg, D. (2024). Case-Based Reasoning to Aid in Clinical Decision Support of TBI. Poster, IEEE Northeast Bioengineering Conference, Hoboken, NJ, April.
- **Okeke, T. K.**, and Sacan, A. (2024). Utilizing Neural Network Autoencoders for Unsupervised Feature Learning from RNA-Seq Differential Expression Analysis Towards Gene Ontology Term Prediction. Poster, Society of Toxicology Annual Meeting, Salt Lake City, UT, March.
- **Okeke, T.**, Siddiq, K., Restrepo, M., Tadikonda, P., and Laudanski, K. (2022). Patterns of Urinary Biomarkers of Immunologic Activation and Sepsis in Patients With COVID-19. *Critical Care Medicine*, 50(1), 74, abstract 181. doi:10.1097/01.ccm.0000807048.74105.59.
- Restrepo, M., **Okeke, T.**, Siddiq, K., Tadikonda, P., and Laudanski, K. (2022). Longitudinal Changes of Neuro-Specific Serum Proteins in COVID-19 Patients. *Critical Care Medicine*, 50(1), 71, abstract 174. doi:10.1097/01.ccm.0000807020.73561.ed.

Upcoming workshop

Primary workshop director and presenter, The 2026 Informatics, Big Data, and AI Workshop, Neurocritical Care Society Annual Meeting, October 20, 2026. With Brandon Foreman, Eric Rosenthal, and Morteza Zabihi.

Honors and awards

- UEV Venture Building Weekend, winning team, Xploit, March 2026.
- Foundry Startup in a Weekend, New Idea prize, Xploit, November 2025. Team award: \$1,550.
- Drexel Startups Fund, Meddibia, June 2024. Team project award: \$20,000.
- Baiada Institute Innovation Tournament, Aboki Learn, February 2024. Team award: \$5,000.
- HESI GTTC Professional Development Award, Spring 2023. Individual award: \$1,000.
- Philly Codefest Collaborative Team Award, Meddibia, March 2023. Team award: \$5,000.
- Drexel University Dean's List, 2019 – 2024.

Leadership

Vice President and Cofounder, Drexel Computational Design, March 2021 – March 2024.